



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

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|---------------|
| JEE(Advanced) |
| PRE-SCORE     |
| 15-10-2021    |

## JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

### PAPER-2

Time : 3 Hours

Maximum Marks : 204

READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

#### QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses** : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant**  $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant**  $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant**  $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum**  $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant**  $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant**  $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum**  $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum**  $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant**  $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

**BEWARE OF NEGATIVE MARKING**

**PART-1 : PHYSICS**

**SECTION-I (i) : (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

*Full Marks* : +3 If **ONLY** the correct option is chosen.

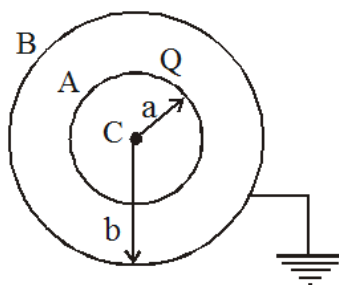
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

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1. Two wires of same material and identical lengths are joined in parallel to a battery of constant voltage. If the radius of cross section of both the wires are in the ratio of 3:5 then the ratio of drift speed of electrons in the wire is given by :  
 (A) 9:25 (B) 5:3  
 (C)  $\sqrt{3} : \sqrt{5}$  (D) None of these
  2. In a vernier callipers 5 vernier scale divisions are equal to 4 main scale divisions. One main scale division is equal to 5 mm. A screw gauge with pitch equal to least count of above vernier callipers is designed. Number of divisions on circular scale of this screw gauge is equal to 100. Least count of this screw is  
 (A) 0.1 mm (B) 0.01 mm (C) 0.001 mm (D) 0.0001 mm
- 

**Space for Rough Work**

3. For a liquid moving in a cylindrical tube of radius  $R$  and length  $\ell$ , the pressure difference at the ends of the tube is related by poiseuille's law given by  $\Delta P = \left( \frac{8\eta\ell}{\pi r^4} \right) Q$ . For a tube having radius  $r = 2.00 \text{ cm} \pm 0.01 \text{ cm}$ ,  $\ell = 10.0 \text{ cm} \pm 0.1 \text{ cm}$  the error in measurement of pressure difference is : (Take :  $\eta = \pi \text{ N-s/m}^2$ ,  $Q = 1 \text{ m}^3/\text{sec}$ )
- (A) 2 %                      (B) 3 %                      (C) 4 %                      (D) 6 %
4. A conducting sphere A of radius  $a$ , with charge  $Q$ , is placed concentrically inside a conducting shell B of radius  $b$ . B is earthed. C is the common centre of the A and B. Then choose the incorrect statement:



- (A) The field at a distance  $r$  from C, where  $a < r < b$  is  $\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
- (B) The potential at a distance  $r$  from C, where  $a < r < b$ , is  $\frac{1}{4\pi\epsilon_0} \frac{Q}{r}$
- (C) The potential difference between A and B is  $\frac{1}{4\pi\epsilon_0} Q \left( \frac{1}{a} - \frac{1}{b} \right)$
- (D) The potential at a distance  $r$  from C, where  $a < r < b$ ,  $\frac{1}{4\pi\epsilon_0} Q \left( \frac{1}{r} - \frac{1}{b} \right)$ .

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Space for Rough Work

**SECTION-I (ii) : (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

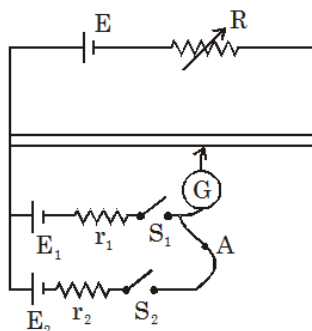
*Negative Marks* : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. In the determination of Young's modulus  $\left( Y = \frac{4MLg}{\pi d^2} \right)$  by using Searle's method, a wire of length  $L = 2$  m and diameter  $d = 0.5$  mm is used. For a load  $M = 2.5$  kg, an extension  $\ell = 0.25$  mm in the length of the wire is observed. Quantities  $d$  and  $\ell$  are measured using a screw gauge and a micrometer, respectively. They have the same pitch of 0.5 mm. The number of divisions on their circular scale is 100. Then mark the incorrect statement(s) regarding contributions to the maximum probable error of the  $Y$  measurement
- (A) due to the errors in the measurements of  $d$  and  $\ell$  are the same.
- (B) due to the error in the measurement of  $d$  is twice that due to the error in the measurement of  $\ell$ .
- (C) due to the error in the measurement of  $\ell$  is twice that due to the error in the measurement of  $d$ .
- (D) due to the error in the measurement of  $d$  is four time that due to the error in the measurement of  $\ell$ .

**Space for Rough Work**

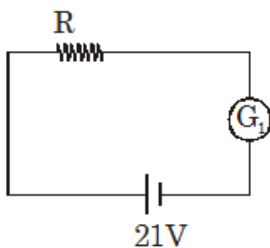
6. Two batteries of EMF  $E_1$  and  $E_2$  and internal resistance  $r_1$  and  $r_2$  respectively are connected across an potentiometer as shown. When only switch  $S_1$  is closed, reading of potentiometer is  $\ell_1$ . When only switch  $S_2$  is closed, reading of potentiometer is  $\ell_2$ . When both switches are closed, reading is  $\ell_3$ . Mark the correct statement(s) :



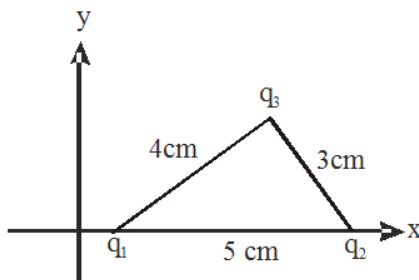
- (A)  $\ell_3 = \frac{\ell_1 + \ell_2}{2}$
- (B)  $\ell_1 < \ell_3 < \ell_2$  or  $\ell_2 < \ell_3 < \ell_1$
- (C) If another galvanometer is attached at point A in series and both switches are closed, it will show zero deflection.
- (D) A change in primary circuit rheostat R changes all the reading of potentiometer.
7. Two infinite, parallel, non-conducting sheets carry equal positive charge density  $\sigma$ . One is placed in the yz plane at  $x = 0$  and the other at distance  $x = a$ . Take potential  $V = 0$  at  $x = 0$ . Choose the correct option(s).
- (A) For  $0 \leq x \leq a$ , potential  $V_x = 0$
- (B) For  $x \geq a$ , potential  $V_x = -\frac{\sigma}{\epsilon_0}(x - a)$
- (C) For  $x \geq a$ , potential  $V_x = \frac{\sigma}{\epsilon_0}(x - a)$
- (D) For  $x \leq 0$  potential  $V_x = \frac{\sigma}{\epsilon_0}x$

Space for Rough Work

8. In the circuit shown, the battery is ideal. It has a voltage of 21 V. The galvanometer  $G_1$  shows 0.42 A. When another identical galvanometer  $G_2$  is connected across R, the reading of  $G_1$  increases to 0.50 A :-



- (A) The resistance of galvanometer is  $30\ \Omega$   
 (B) The resistance  $R = 40\ \Omega$   
 (C) The reading of  $G_2 = 0.20\ \text{A}$   
 (D) If instead of this,  $G_2$  was connected parallel to  $G_1$ , their reading would be same = 0.21A.
9. Three charges  $q_1$ ,  $q_2$  &  $q_3$  are placed as shown. The magnitude of ' $q_1$ ' is  $2\mu\text{C}$ , but its sign and the value of charge ' $q_2$ ' are not known. Charge ' $q_3$ ' is  $+4\mu\text{C}$ , and net force  $\vec{F}$  on  $q_3$  is in the negative 'x' direction.

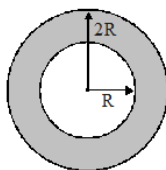


- (A) Charge ' $q_1$ ' is negative.  
 (B) Charge ' $q_2$ ' is positive  
 (C) Magnitude of charge ' $q_2$ ' is  $\frac{27}{32}\mu\text{C}$ .  
 (D) The direction of net force on ' $q_1$ ' cannot be determined.

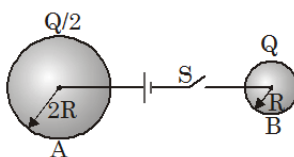
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Space for Rough Work

10. Consider a spherical volume which has uniform charge density  $\rho$  from  $x = R$  to  $x = 2R$ . There is no charge from  $x = 0$  to  $x = R$  and from  $x = 2R$  to infinity. The electric potential due to this charge :



- (A) at  $x = 0$  is 0  
 (B) at  $x = \frac{R}{2}$  is  $\frac{3\rho R^2}{2\epsilon_0}$   
 (C) at  $x = 2R$  is  $\frac{7\rho R^2}{6\epsilon_0}$   
 (D) at  $x = \frac{3R}{2}$  is  $\frac{8\rho R^2}{3\epsilon_0}$
11. Two metallic spheres of radius  $2R$  and  $R$  placed far away are charged to  $Q/2$  and  $Q$  respectively and connected by a battery of emf  $V$  as shown in figure. When switch  $S$  is closed :



- (A) potential difference between these two sphere will not be equal to  $V$  because they are already charged  
 (B) charges will redistribute until both spheres will have equal potential  
 (C) charge on sphere A will increase whereas charge on sphere B will decrease  
 (D) total charge on sphere A and B will still remain  $3Q/2$
12. Two resistances are measured in Ohm.  
 $R_1 = 3\Omega \pm 1\%$ ;  $R_2 = 6\Omega \pm 2\%$
- (A) When they are connected in parallel, the percentage error in equivalent resistance is 4.67%  
 (B) When they are connected in parallel, the percentage error in equivalent resistance is 0.67%  
 (C) When they are connected in series, the percentage error in equivalent resistance is 1.66%  
 (D) When they are connected in series, the percentage error in equivalent resistance is 5%

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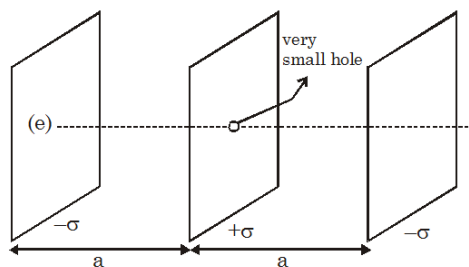
Space for Rough Work

SECTION-I (iii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case)

Paragraph for Questions 13 and 14

Three parallel infinite nonconducting uniformly charged (charge density  $\sigma$ ) plates are arranged as shown in figure. A small hole passes through middle plate. At  $t = 0$  an electron (charge  $e$ , mass  $m$ ), is released from left most plate with zero initial velocity along the axis of hole.



13. Velocity of electron as it crosses the hole is :-

(A)  $\sqrt{\frac{e\sigma a}{2m\epsilon_0}}$  (B)  $\sqrt{\frac{e\sigma a}{m\epsilon_0}}$  (C)  $\sqrt{\frac{2e\sigma a}{m\epsilon_0}}$  (D)  $2\sqrt{\frac{e\sigma a}{m\epsilon_0}}$

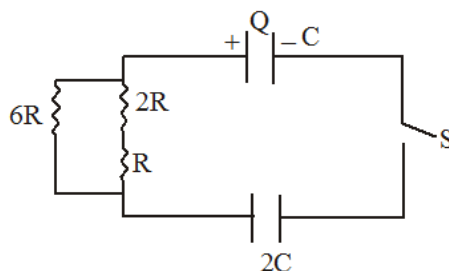
14. At  $t = 0$  an electron (charge  $e$ , mass  $m$ ) emerges from the hole moving perpendicular to the plates with speed  $v_0$ . Assuming  $v_0$  is small enough that the electron does not collide with the negative plate. Find the period of its periodic motion :-

(A)  $\frac{4v_0 m\epsilon_0}{e\sigma}$  (B)  $\frac{8v_0 m\epsilon_0}{e\sigma}$  (C)  $\frac{2v_0 m\epsilon_0}{e\sigma}$  (D)  $\frac{v_0 m\epsilon_0}{e\sigma}$

Space for Rough Work

Paragraph for Questions 15 and 16

In the given figure switch 'S' is closed at  $t = 0$



15. Mark the correct option(s) :

- (A) Charge on capacitor  $2C$  as function of time is  $\frac{2Q}{3} \left(1 - e^{-\frac{3t}{4RC}}\right)$ .
- (B) Charge on capacitor  $C$  as function of time is  $\frac{Q}{3} \left(1 - e^{-\frac{3t}{4RC}}\right)$ .
- (C) Current through the resistance  $6R$  as a function of time is  $\frac{Q}{2RC} e^{-\frac{3t}{4RC}}$ .
- (D) Current through the resistance  $6R$  as a function of time is  $\frac{Q}{6RC} e^{-\frac{3t}{4RC}}$ .

16. Mark the correct option(s) :

- (A) Amount of heat produced in resistance ' $R$ ' in long time is  $\frac{2Q^2}{27C}$ .
- (B) Amount of heat produced in resistance ' $2R$ ' in long time is  $\frac{4Q^2}{27C}$ .
- (C) Amount of heat produced in resistance ' $6R$ ' in long time is  $\frac{Q^2}{9C}$ .
- (D) Amount of heat produced in resistance ' $6R$ ' in long time is  $\frac{Q^2}{18C}$ .

Space for Rough Work

**Paragraph for Questions 17 and 18**

The universal gravitational constant is measured experimentally using Cavendish's experiment. This experiment is a very precise experiment which gives the value of  $G$  as  $(6.6742 \pm 0.0010) \times 10^{-11} \text{ N.m}^2/\text{kg}^2$ . Since the universal gravitational constant has some error in it, using it if we find escape speed and acceleration due to gravity on Earth's surface, our values will have error in it. (Assume mass of Earth =  $6.0000 \times 10^{24} \text{ kg}$  & radius of Earth =  $6.4000 \times 10^6 \text{ m}$ ). The measurement of mass of Earth has an error of 0.01% and radius of Earth has an error of 0.02%.

17. Choose the correct option(s) :
- (A) Maximum percentage error in acceleration due to gravity  $g = 0.065$
  - (B) Maximum percentage error in acceleration due to gravity  $g = 0.025$
  - (C) value of acceleration due to gravity  $g$ (approx. in  $\text{m/s}^2$ ) = 9.78
  - (D) value of acceleration due to gravity  $g$ (approx. in  $\text{m/s}^2$ ) = 8.78
18. What will be the maximum percentage error in calculating the escape speed on earth.
- (A) 0.045%
  - (B) 0.0225%
  - (C) 0.015%
  - (D) 0.0025%

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**Space for Rough Work**

**PART-2 : CHEMISTRY**  
**SECTION-I (i) : (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

1. Two liquids A and B have,  $P_A^0$  and  $P_B^0 = 1 : 3$  at a certain temperature. Assume A and B form an ideal solution, and the ratio of mole of A to B in the liquid phase are 1 : 3, then mole fraction of A in vapour phase in equilibrium with the solution -  
 (A) 0.1 (B) 0.2 (C) 0.5 (D) 1.0
2.  $\text{Mg} + \underset{\substack{\text{elemental solid} \\ \text{having atomicity 12}}}{\text{Black}} \xrightarrow{\Delta} \text{'X'} \xrightarrow{\text{H}_2\text{O}} \text{'Y'} \uparrow + \dots + \dots$   
 $\text{'Y'} + \text{CO} \longrightarrow \text{'Z'}$   
 which of the following statement is correct?  
 (A) Maximum number of atoms are lying in one plane is 4 for 'Z'.  
 (B) Y has no electron deficiency.  
 (C) 'X' is a covalent compound.  
 (D) Y has eight 2c-2e bond.

Space for Rough Work

3. Which of the following gives Isocyanide test with  $\text{CHCl}_3 + \text{KOH}$  :
- (A) N-methylcyclohexanamine  
(B) Acetamide  
(C) Aniline  
(D) N,N-dimethylaniline
4. Sweetest artificial sweetner is :
- (A) Alitame  
(B) Saccharin  
(C) Sucralose  
(D) Aspartame

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**Space for Rough Work**

**SECTION-I (ii) : (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

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*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

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5. The co-ordination number of FCC structure for metals is 12, since
- (A) each atom touches 4 others in same layer, 3 in layer above and 3 in layer below.
- (B) each atom touches 4 others in same layer, 4 in layer above and 4 in layer below.
- (C) each atom touches 6 others in same layer, 3 in layer above and 3 in layer below.
- (D) each atom touches 3 others in same layer, 6 in layer above and 6 in layer below.
6. Select the correct information(s) regarding truncated tetrahedron
- (A) 12 vertices (B) 8 faces
- (C) 18 edges (D) 3 hexagonal faces
- 

**Space for Rough Work**

7. If a thin slice of sugar beet is placed in concentrated solution of NaCl then
- (A) Sugar beet will lose water from its cells
  - (B) Sugar beet will absorb water from solution
  - (C) Sugar beet will neither absorb nor lose water
  - (D) Sugar beet will dissolve in solution
8. Consider hydrogen halides (HX) and select correct statement(s).
- (A) Dissolves in water and produces hydrohalic acid
  - (B) Dissolves in water and produces halous acid
  - (C) Most heavier hydrogen halide is most acidic in water
  - (D) Lightest hydrogen halide has maximum boiling point
9. Select correct statement(s)?
- (A) Borax is prepared from colemanite
  - (B) Boric acid is produced by reaction of borax with conc. HCl
  - (C) Diborane is prepared by heating boric acid
  - (D) Boric acid is produced by hydrolysis of diborane
10.  $\text{Cl}_2$  gas can be produced by which of the following process/es –
- (A) Electrolysis of fused NaCl
  - (B) Acidification of Bleaching powder
  - (C) Reaction of conc. HCl with  $\text{KMnO}_4(\text{aq.})$
  - (D) Heating KCl with conc.  $\text{H}_2\text{SO}_4$

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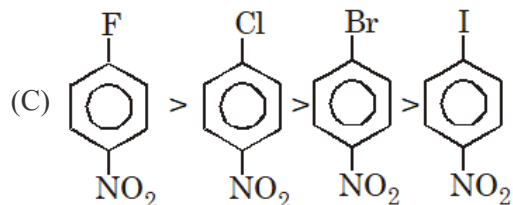
**Space for Rough Work**

11. Select correct statement(s)?

(A) HVZ reaction is used for replacement of  $\alpha$ -hydrogen in carboxylic acid by halogen

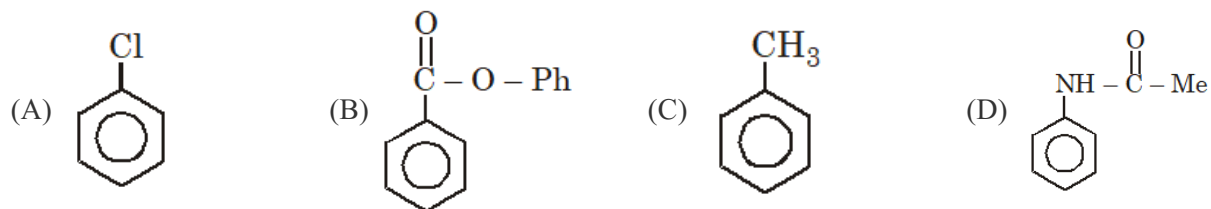
(B) Gabriel's phthalimide synthesis is best method to prepare  $\text{Me}_3\text{C-NH}_2$

Correct order of reactivity for reaction with aq. NaOH



(D) Glycine, glutamic acid and lysine can be separated by electrophoresis.

12. Which of the following will react at rate faster than benzene during aromatic electrophilic substitution?



Space for Rough Work

SECTION-I (iii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case)

Paragraph for Questions 13 and 14

Spinel is an important class of oxide having two types of metal ions. A type of spinel have ccp arrangement to  $O^{2-}$  ions in which  $Fe^{2+}$  cation occupy  $1/8^{th}$  of tetrahedral voids and  $Fe^{3+}$  cation occupies half of the octahedral voids. If 'a' be the edge of unit cell then.

13. What is formula of spinel -

- (A)  $Fe_2O_3$  (B)  $Fe_3O_4$  (C)  $FeO$  (D)  $Fe_4O_5$

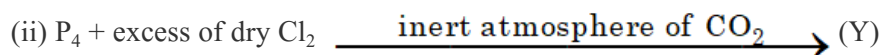
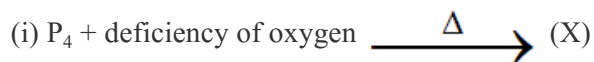
14. What is the minimum internuclear distance between  $Fe^{2+}$  ion &  $Fe^{3+}$  ion in spinel -

- (A)  $\frac{\sqrt{3}a}{4}$  (B)  $\frac{a}{2}$  (C)  $\frac{a}{\sqrt{2}}$  (D) a

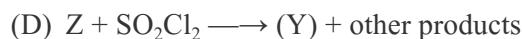
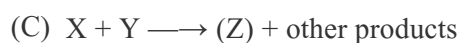
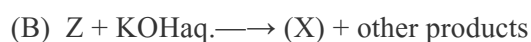
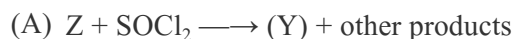
Space for Rough Work

**Paragraph for Questions 15 and 16**

Identify the principal phosphorous containing products X, Y and Z for each of the following reactions and give answer of the given questions.



15. Select incorrect reaction(s) ?



16. Finely divided Sn on heating with Y produces –



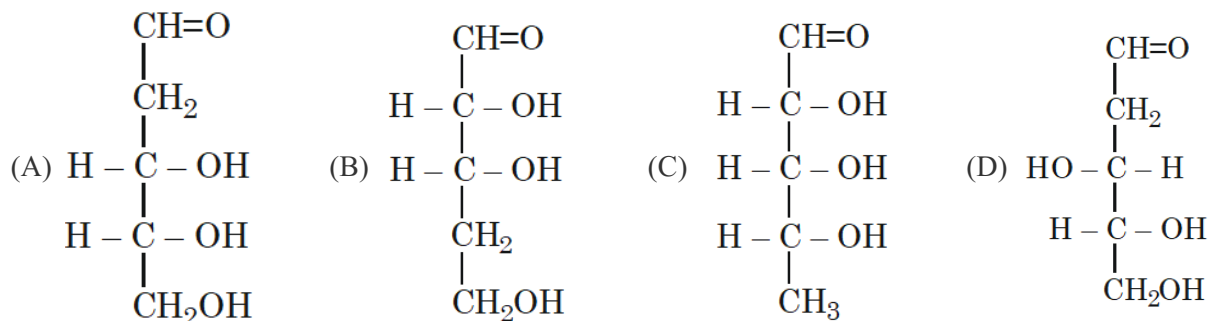

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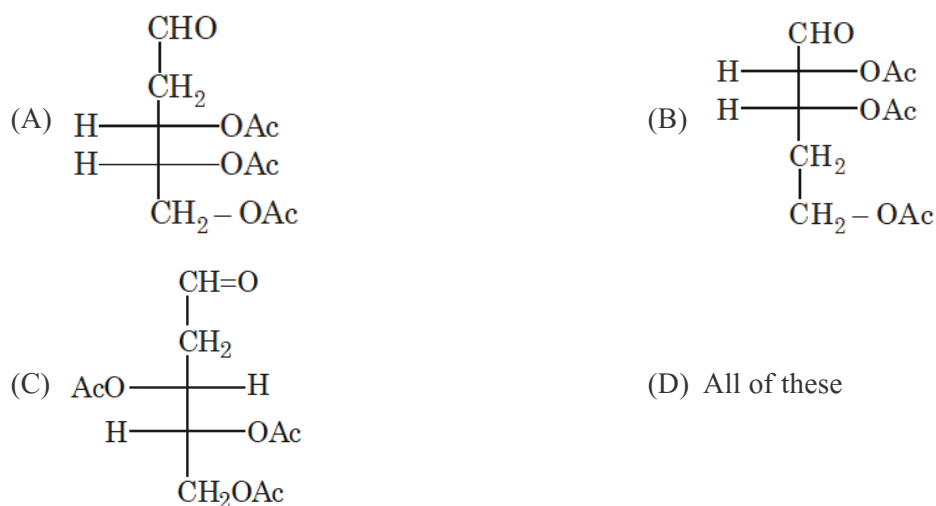
Paragraph for Questions 17 and 18

Compound A ( $C_5H_{10}O_4$ ) on oxidation with  $Br_2 + H_2O$  gives acid B ( $C_5H_{10}O_5$ ). A forms triacetate B by acetic anhydride and is reduced by red P + HI to give n-pentane. Oxidation of A with  $HIO_4$  gives among other products, 1 mole of  $H_2C=O$  & 1 mole of  $HCO_2H$ .

17. The possible structure of A can be :



18. B can be :



Space for Rough Work

**PART-3 : MATHEMATICS**  
**SECTION-I (i) : (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

1. If a function  $f : [-2, \infty) \rightarrow \mathbb{R}$  is such that  $f(x) = x^2 + 4x - |x^2 - 4|$ , then range of  $f(x)$  is -  
 (A)  $(-\infty, \infty)$  (B)  $[-6, 12]$  (C)  $[-6, \infty)$  (D)  $[-6, 4]$
2. Consider  $\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$ . If  $\alpha, \beta$  and  $\gamma$  be the projection of  $\vec{r}$  along  $\frac{2\hat{i} + \hat{j}}{\sqrt{5}}, \frac{-2\hat{i} + \hat{j}}{\sqrt{5}}$  and  $\hat{k}$  respectively such that  $|\alpha| = |\beta| = |\gamma|$  and  $|\vec{r}| = 1$ , then number of such ordered triplets (a, b, c) is  
 (A) 6 (B) 7 (C) 8 (D) 9
3. If  $x = \alpha$  satisfies the equation  $\frac{\sin^{-1}(x^{2018}) + 2\cos^{-1}(x^{2017})}{\cos^{-1}(x^{2018}) + 2\sin^{-1}(x^{2017})} = \frac{-5}{2}$   
 then the value of  $\frac{\sin^{-1}(\sin \alpha) + \tan^{-1}(\tan \alpha)}{\cos^{-1}(\cos \alpha) + 2 \operatorname{cosec}^{-1}(\operatorname{cosec} \alpha)}$  is  
 (A) 1 (B) 2 (C) 3 (D) 4
4. Number of integer(s) in the range of  $f(x) = \tan^{-1} \frac{|x|}{\sqrt{1-x^2}} + \cos^{-1} \frac{1}{\sqrt{1+x^2}}$   
 (A) 2 (B) 3 (C) 4 (D) 5

Space for Rough Work

**SECTION-I (ii) : (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. The function defined by  $f(x) = \frac{ax+b}{cx+d}$ , where a, b, c, d are non zero real numbers,  $f(19) = 19$ ,  $f(97) = 97$ ,  $f(f(x)) = x$  for all values except  $-\frac{d}{c}$ . The number that is **NOT** in the range of f(x) is
- (A) 57 (B) 58 (C) 59 (D) 60
6. Let (x,y) denote a solution of equation  $\tan^{-1}x + \cot^{-1}y = \sin^{-1}\frac{3}{\sqrt{10}}$ , where x, y  $\in \mathbb{N}$ . Value of (x + y) can be -
- (A) 3 (B) 6 (C) 8 (D) 9

**Space for Rough Work**

7. The vector  $\hat{i} + x\hat{j} + 3\hat{k}$  is rotated about its initial point through an angle of  $\cos^{-1} \frac{11}{14}$  and its magnitude is doubled. If the vector in the new position is given by  $4\hat{i} + (4x - 2)\hat{j} + 2\hat{k}$ . Then which of the following can't be the value of  $x$  ?
- (A)  $-\frac{2}{3}$  (B)  $\frac{2}{3}$  (C)  $-\frac{20}{17}$  (D) 2
8. If  $(\vec{\alpha} \times \vec{\beta}) \times (\vec{\gamma} \times \vec{\alpha}) + \vec{\alpha} + \vec{\beta} = \vec{\beta} \times \vec{\gamma}$  where  $\vec{\alpha}$  and  $\vec{\beta}$  are unit vectors, then choose the correct option(s)
- (A)  $\vec{\gamma}$  is parallel to  $\vec{\alpha}$  (B)  $\vec{\gamma}$  is parallel to  $\vec{\beta}$
- (C) there is exactly one value of  $|\vec{\gamma}|$  (D) there are infinite values of  $|\vec{\gamma}|$
9. Let  $P(r)$  denote number of points  $(x, y)$  (where  $x, y \in I$ ) which lie inside or on the boundary of the region enclosed by the curve whose joint equation is  $|xy| - r(|x| + |y|) + r^2 = 0$ ,  $r \in \mathbb{N}$  and if  $\lim_{n \rightarrow \infty} \frac{\left( \sum_{r=1}^{r=n} P(r) \right) - \lambda n^3}{an^2 + bn + c} = \frac{1}{2}$  then which of the following is/are true
- (A)  $\lambda = \frac{4}{3}$  (B)  $a = 4$  (C)  $a = 8$  (D)  $\lambda = \frac{2}{3}$

Space for Rough Work

10. Let  $f : \mathbb{R} \rightarrow (0, 1)$  be a continuous function. Then which of the following functions must have the value zero at some point in the interval  $(0, 1)$ ?

(A)  $x^7 - f(x)$  (B)  $x^2 - \int_0^x f(t) dt$

(C)  $\int_0^{\frac{\pi}{2}-x} f(t) \sin 2t dt - x$  (D)  $\sqrt{x} - \int_0^{x^2} \sqrt{f(t)} dt$

11. If  $f(x) : [-1, 4] \rightarrow [-1, 2]$  is an onto function, then which following is/are correct -

(A) Domain of  $f(|x| - 1)$  is  $[-5, 5]$  (B) Range of  $f(x) + 1$  is  $[0, 3]$

(C) Range of  $f(x + 1)$  is  $(-2, 2)$  (D) Domain of  $f(|x|)$  is  $[-4, 4]$

12. Consider  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 2([x] - \{x\})$  and  $g : \mathbb{R} \rightarrow \mathbb{R}$ ,  $g(x) = 2 \left[ \frac{x}{2} \right] - \frac{x}{2}$ , then which of the following holds good ?

(where  $[.]$  and  $\{.\}$  denotes greatest integral function & fractional part function respectively)

- (A) Area of triangle formed by  $y = f \circ g(x)$ ,  $x = 0$ ,  $y = 0$  is 2.
- (B) Area of triangle formed by  $y = g \circ f(x)$ ,  $x = 0$ ,  $y = 0$  is 2.
- (C)  $f(x)$  is bijective function.
- (D)  $g(x)$  is bijective function.

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Space for Rough Work

**SECTION-I (iii) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:  
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*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case)

**Paragraph for Questions 13 and 14**

A plane P contains the line  $L_1 : \frac{y}{b} + \frac{z}{c} = 1, x = 0$  and is parallel to the line  $L_2 : \frac{x}{a} - \frac{z}{c} = 1, y = 0$

13. If the shortest distance between  $L_1$  and  $L_2$  is then the value of  $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$  is completely divisible by :
- (A) 16 (B) 64 (C) 128 (D) 192
14. Distance of image of  $A(a, 0, 0)$  in the plane P from  $M\left(-\frac{5}{3}, \frac{8}{3}, \frac{11}{3}\right)$ , where  $a = b = c = 1$  is equal to :
- (A) 1 (B) 2 (C) 3 (D) 4

**Space for Rough Work**

**Paragraph for Questions 15 and 16**

ABCD is a regular tetrahedron.  $M(0, 0, 0)$  is the midpoint of the altitude DN of the tetrahedron. Coordinates of the points A and B are  $(0, 0, -2)$  and  $(b, b, 0)$  respectively (with  $b > 0$ ). Also x-coordinate of the point C is given to be positive

15. Image of the point D in the plane generated by ABC is

(A)  $(2\sqrt{2}, 0, -2)$

(B)  $(2, 0, -2)$

(C)  $(2, 0, -2\sqrt{2})$

(D) None of these

16. Volume of the tetrahedron ACBM is

(A)  $\frac{5\sqrt{2}}{6}$

(B)  $\frac{4\sqrt{2}}{3}$

(C)  $\frac{4}{3}$

(D) None of these

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**Space for Rough Work**

**Paragraph for Questions 17 and 18**

Let  $f : [0, \infty) \rightarrow [2, \infty)$  be a differentiable increasing and onto function satisfying  $f^2(x) - 2f(x) - \sqrt{x} = f^2(y) - 2f(y) - \sqrt{y}$ .

On the basis of above information, answer the following questions :

17. The value of  $\lim_{x \rightarrow 0^+} \frac{f(x) - 2}{\sqrt{x}}$  is not equal to -
- (A)  $\frac{1}{\sqrt{2}}$  (B)  $\frac{1}{2}$
- (C)  $\frac{1}{3}$  (D) 0
18. If  $g(x)$  be the inverse of  $f(x)$ , then  $g'(4)$  is equal to -
- (A) 16 (B)  $\frac{1}{16}$
- (C) 96 (D)  $\frac{1}{96}$

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**Space for Rough Work**

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Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

Space for Rough Work